Minds on the Wall: A Critical Evaluation of Conceptual Blending and Metaphor Theory for Interpreting European Ice Age Imagery

I. Introduction: The Cognitive Chasm and the Symbolic Bridge

The study of European Ice Age art confronts the modern researcher with a profound cognitive and cultural chasm. Separated by over a thousand generations, we are left with a silent record of paintings, engravings, and sculptures, compelling in their beauty but enigmatic in their purpose.[1] These artifacts are more than mere decoration; they represent a crucial archaeological trace of complex cognitive processes, a veritable "fossil record of the mind".[2] The central challenge lies in bridging this temporal gap without succumbing to speculative fantasy. Traditional interpretive frameworks, while valuable, have often highlighted the underdetermined nature of the evidence, leading to a proliferation of competing theories.[4] This report evaluates the potential of two powerful theoretical tools from cognitive linguistics—Conceptual Metaphor Theory (CMT) and Conceptual Integration (Blending) Theory—to serve as a more rigorous bridge across this chasm. It posits that while these theories offer an unparalleled framework for modeling the cognitive architecture behind Paleolithic imagery, their application must be tempered by a rigorous critique of their universalist assumptions. By examining cultural variations in metaphor and the peril of anachronism, their use can be refined from a tool for "decoding" static meaning to a more powerful heuristic for generating testable hypotheses about the processes of meaning-making in the deep past.

The Environmental Context of the Ice Age Mind

To understand these cognitive acts, one must first situate them within their proper environmental context. The relevant period is not the recent and comparatively mild "Little Ice Age" of the last millennium [5], but the far more severe climatic conditions of the Upper Paleolithic, particularly during and after the Last Glacial Maximum (LGM), an intense cold period from approximately 27,000 to 19,000 years ago.[8] This was an era of extreme environmental stress, characterized by the dramatic expansion of continental ice sheets that covered most of northern and central Europe.[9] This severe cooling had a major impact on all life, driving human populations into southern refugia and causing significant demographic fluctuations.[8] The human population of Europe may have contracted from over 300,000 individuals before the LGM to as few as 130,000 at its peak.[8] This context of survival, resource scarcity, and population bottlenecks is crucial. The "creative explosion" of symbolic behavior seen in the art of this period should not be viewed as a leisurely artistic pursuit, but as a potential cognitive adaptation for enhancing social cohesion, storing and transmitting vital ecological information, and navigating a high-stress world.[2] The art, in this light, becomes a key element of a sophisticated survival strategy.

The Interpretive Challenge

The fundamental difficulty in studying Paleolithic art is the profound absence of direct cultural context.[10] Lacking written records or direct continuity with the creators, researchers are forced to infer meaning from material patterns alone. This process is fraught with the danger of anachronism—the projection of our own cultural biases, assumptions, and conceptual frameworks onto a distant past.[1] Over the last century, this has led to a succession of interpretive paradigms. Early theories proposed that the art served a utilitarian function of "sympathetic magic" to ensure success in the hunt.[4] Later, structuralist approaches, pioneered by André Leroi-Gourhan, focused on perceived spatial patterns and dualisms within the caves, suggesting a complex, organized symbolic system.[15] More recently, neuropsychological models championed by David Lewis-Williams have interpreted much of the art, particularly its geometric signs and therianthropic (human-animal) figures, as representations of shamanic visions experienced in altered states of consciousness.[4]

The very existence of these multiple, competing, and often mutually exclusive interpretations is not a sign of scholarly failure. Rather, it is a diagnostic indicator of the nature of the problem itself. It demonstrates that the art is polysemous and that any purely content-based interpretation is inherently underdetermined by the available evidence. The data can support multiple "readings," making it difficult to falsify any single claim about what the art "meant."

This interpretive impasse justifies a methodological pivot.

Cognitive Archaeology as a New Path

Cognitive archaeology offers such a pivot by changing the fundamental question. Instead of asking the perhaps unanswerable question, "What did this art *mean*?", it asks, "What cognitive capacities were necessary to *make* this art?".[3] This approach shifts the focus from static, decoded meaning to the dynamic processes of meaning-making. It analyzes artifacts as tangible evidence for the evolution of specific cognitive faculties, such as abstract thought, symbolic reasoning, working memory, and the ability to conceive of non-existent entities.[17] By applying formal models from the cognitive sciences, this discipline seeks to build a more rigorous, constrained, and testable framework for understanding the minds of our ancestors.[19] It is within this framework that Conceptual Metaphor and Blending Theory find their application, offering a detailed language for describing the mental operations that may have given rise to the images on the walls.

II. The Architecture of Modern Thought: Conceptual Metaphor and Integration Theories

To evaluate the utility of these theories for Paleolithic art, it is first necessary to provide a detailed exposition of their core principles. These are not merely theories of language but comprehensive models of human cognition, proposing that our capacity for abstract thought is fundamentally grounded in our bodily experience.

2.1. Metaphors We Lived By: Conceptual Metaphor Theory (CMT)

Pioneered by George Lakoff and Mark Johnson, Conceptual Metaphor Theory (CMT) revolutionized the study of metaphor by arguing that it is not a decorative linguistic device but a fundamental cognitive mechanism.[21] The essence of CMT is that we understand and experience one kind of thing, typically an abstract concept (the *target domain*), in terms of another, more concrete concept (the *source domain*).[23] This cognitive process is directional, systematically mapping knowledge from the concrete and experiential to the abstract and

intangible.[24] For example, in the conceptual metaphor , the abstract target domain of is structured by the concrete source domain of a . This is not a single expression but a system of correspondences, or mappings: life has a start and end, people encounter obstacles, choose paths, and have companions.[23] Similarly, in , we *attack* an opponent's weak points, *defend* our own position, *win* or *lose* debates, and *shoot down* flawed ideas.[25]

The entire edifice of metaphorical thought is, according to CMT, grounded in our embodied experience of the world. This grounding is achieved through two foundational cognitive structures:

- **Image Schemas:** These are recurring, dynamic, pre-conceptual patterns that emerge from our sensorimotor interactions.[26] They are not rich mental images but skeletal structures of experience, such as (in/out), (movement from one point to another), , and (verticality).[23] These schemas are considered pre-linguistic and possibly universal, arising from our shared human embodiment and interaction with a physical environment.[30]
- **Primary Metaphors:** These are the "atomic" components of metaphorical thought, formed by the repeated co-occurrence, or "conflation," of a subjective experience with a sensorimotor experience in early life.[32] For example, the experience of adding more fluid to a container and seeing the level rise leads to a neural association between quantity and verticality, forming the primary metaphor .[32] Similarly, the experience of physical warmth from a caregiver being correlated with feelings of affection gives rise to .[23] Hundreds of such primary metaphors (e.g., , ,) form the basic vocabulary of abstract thought.[23] More complex, or "molecular," metaphors like are then constructed by combining these primary components.[23]

This hierarchical structure, progressing from basic bodily schemas to primary metaphors and then to complex metaphorical systems, provides a cognitive "supply chain" for abstract thought. This progression offers a powerful analytical framework for archaeology. It suggests a principle of parsimony: one can search for evidence of the simplest components (e.g., depictions reflecting a or schema) before positing the existence of highly complex, culturally specific metaphorical concepts, allowing for a more rigorous, bottom-up reconstruction of cognitive capacities.

2.2. The Way We Think: Conceptual Integration (Blending) Theory

Developed by Gilles Fauconnier and Mark Turner, Conceptual Integration Theory, or Blending Theory, offers a more general and dynamic model of how new meaning is constructed in real-time.[38] It describes a cognitive process where elements from different "mental spaces"—small, temporary conceptual packets built as we think and talk—are combined to

create novel ideas.[39]

The basic integration network consists of four spaces [38]:

1. Two (or more) **Input Spaces**, which contain the distinct conceptual structures to be blended.
2. A **Generic Space**, which captures the structure, roles, or schema that the input spaces have in common.
3. A **Blended Space (or "Blend")**, into which selected elements from the inputs are projected. Within this space, these elements are combined through processes of *composition*, *completion*, and *elaboration*, leading to an *emergent structure*—new meaning that is not present in any of the inputs alone.

Blending Theory is a powerful model for creativity, analogy, and counterfactual reasoning. While it can model the operations underlying many conceptual metaphors, its most significant contribution is the concept of **Double-Scope Blending**. This is the most complex and arguably uniquely human form of conceptual integration.[42] It occurs when the two input spaces have different, and often conflicting, organizing frames or structures. The blend selectively imports partial structure from *both* inputs and develops its own novel, emergent frame.[43] This allows for the creation of concepts that are impossible in the real world but are conceptually coherent, such as a "computer desktop" (blending the frames of a physical office desk and a digital file system) or a "talking animal" in a fable.[2]

This concept of double-scope blending provides a formal cognitive mechanism for what archaeologist Steven Mithen has termed "cognitive fluidity".[44] In his influential work, Mithen proposed that the modern human mind emerged around 50,000 years ago when previously isolated, domain-specific intelligences—such as a 'natural history intelligence' (for understanding animals and plants), a 'social intelligence' (for navigating group dynamics), and a 'technical intelligence' (for tool-making)—began to flow into one another.[2] Double-scope blending theory precisely models this process at the level of individual concepts. The 'human' domain and the 'animal' domain, for example, represent two input spaces with fundamentally conflicting organizing frames (e.g., a person has kinship and uses tools; a lion belongs to a pride and hunts with claws). The integration of these two frames into a single blended space is an act of double-scope blending, resulting in a novel, hybrid concept like a therianthrope. Thus, Blending Theory provides the operational mechanics for Mithen's proposed architectural change in the human mind, explaining *how* cognitive fluidity works.

Table 1: Comparative Analysis of Conceptual Metaphor Theory and Conceptual Integration Theory

Feature	Conceptual Metaphor	Conceptual Integration

	Theory (CMT)	(Blending) Theory
Core Idea	Understanding one (abstract) domain in terms of another (concrete) one.	Creating new meaning through the dynamic combination of multiple mental inputs.
Basic Units	Source Domains, Target Domains.	Mental Spaces (Input, Generic, Blend).
Key Mechanism	Systematic, directional mapping of structure from source to target.	Selective projection, composition, completion, and elaboration in the blend.
Primary Output	A stable, conventionalized system of metaphorical expressions.	An emergent structure in the blend, often created "online" for local purposes.
Relation to Embodiment	Fundamentally grounded in embodied experience via image schemas and primary metaphors.	A general cognitive operation that can operate on any inputs, including embodied ones.
Classic Example		The "Computer Desktop"

III. Minds Made Material: Applying Cognitive Theories to Ice Age Therianthropes

Therianthropic figures, which combine human and animal features, are among the most compelling images from the Paleolithic. Because they depict beings that do not exist in the physical world, they are undeniable evidence of abstract, imaginative thought.[47] As such, they are prime candidates for analysis using the frameworks of CMT and Blending Theory.

3.1. Case Study: The *Löwenmensch* of Hohlenstein-Stadel as a

Double-Scope Blend

The *Löwenmensch* ("Lion-man") figurine, discovered in the Hohlenstein-Stadel cave in Germany, is a masterpiece of Aurignacian art, dated to approximately 40,000 years ago.[49] Carved from a mammoth tusk, the 31-centimeter sculpture required an immense investment of time and skill; experimental replication suggests a creation time of over 400 hours, a significant commitment for a small hunter-gatherer community.[48] This effort underscores its profound cultural importance. As one of the earliest and most complex representations of a supernatural being, it marks a major milestone in human cognitive evolution.[47]

From a cognitive perspective, the *Löwenmensch* is a stunning material manifestation of a double-scope conceptual blend.[2] The cognitive operation can be deconstructed as follows:

- **Input Space 1 (HUMAN):** This mental space contains the organizing frame of a human being, including concepts like an upright, bipedal stance, arms for manipulation, complex social roles, tool use, and human kinship structures.
- **Input Space 2 (CAVE LION):** This space contains the conflicting frame of a powerful predator. Key concepts include quadrupedal locomotion, apex predator status in the Ice Age ecosystem, immense strength, ferocity, and animal social structures (e.g., a pride).[48]
- **Generic Space:** This space captures the abstract structure common to both inputs: a concept of a powerful, animate, sentient agent that exists and acts in the world.
- **The Blend (LION-MAN):** Through selective projection, the artist combines the upright bipedal stance and general body plan from the HUMAN input with the head, powerful shoulders, and leonine gaze from the CAVE LION input.[48] The result is an emergent structure: a new, supernatural entity that is neither fully human nor fully lion. This blended being possesses qualities of both—the agency and posture of a human, the power and ferocity of a lion—and something more, a transcendent quality that is unavailable in either input space alone.[47] It embodies a novel concept, perhaps of transformation, spiritual power, or a unique link between the human and animal worlds, demonstrating a capacity for abstract imagination that is a hallmark of the modern human mind.[53]

The sheer effort invested in carving the *Löwenmensch* points to a deeper cognitive function. A complex and novel double-scope blend is an effortful and potentially fleeting mental construction.[53] By externalizing this abstract concept into a durable, physical object, the creator reified it. The figurine becomes a stable, shareable "cognitive artifact" that can anchor collective thought, narrative, and ritual.[48] It offloads the cognitive burden of maintaining the complex concept purely within individual minds, transforming a private thought into a public symbol. In this sense, the *Löwenmensch* acts as an external component of a distributed cognitive system, bridging Blending Theory with theories of extended and material cognition.[54]

3.2. Case Study: Deconstructing "The Sorcerer" of Les Trois Frères

Dated to the later Magdalenian period, around 13,000 BCE, the figure known as "The Sorcerer" is a painted and engraved composite image located high on a wall in a deep, difficult-to-access chamber of the Les Trois Frères cave in the French Pyrenees, a space aptly named 'The Sanctuary'.[57] The figure is a complex amalgam, combining features from a multitude of animals—the antlers of a stag, the round eyes of an owl, what have been described as bear-like paws, and the tail of a horse or wolf—with the legs, posture, and genitalia of a human male.[58] Its dominant position overlooking a throng of engraved animals has led to interpretations of it as a shaman, a deity, or a "master of animals" who presides over the hunt.[57]

Cognitively, "The Sorcerer" represents a more complex and arguably different type of integration network than the *Löwenmensch*. It is a multi-input blend that draws selectively from a wide range of source domains:

- **Multiple Input Spaces:** The blend is not a simple fusion of two frames but a dynamic collage drawing from numerous inputs: , , , , , etc.
- **Metaphorical Entailments:** The blend is conceptual, not merely visual. By combining these inputs, the artist creates a figure that metaphorically entails the salient attributes of each source. The stag's antlers might be mapped to concepts of virility, dominance, or a connection to the forest. The owl's forward-facing eyes could entail nocturnal vision, wisdom, or silent predation. The bear's paws could entail strength. The human posture implies agency, intention, and ritual action.
- **Emergent Narrative:** The emergent structure in this blend is a narrative concept. It is not just a creature, but a role: a being who presides over the animal world, who can traverse the boundaries between species, or who embodies the collected powers of the most significant animals in the environment. This aligns powerfully with shamanic interpretations, where transformation and mediation between the human world and the spirit world are key functions of the ritual specialist.[4] The figure is not just a sum of its parts; it is a new conceptual role created by their integration.

The visual difference between the *Löwenmensch* and "The Sorcerer" may reflect a difference in the *type* of conceptual blend being represented. The *Löwenmensch* is a coherent, seamless fusion of two beings, suggesting the creation of a new, stable ontological category: a lion-person, a specific type of being. "The Sorcerer," in contrast, is a more collage-like assembly of distinct parts from many different beings. This may represent a more dynamic, transient state of transformation—a process rather than a state of being. It could depict a shaman *in the act* of assuming the powers and forms of multiple animal spirits, where various identities are temporarily held by a single agent. This suggests that Blending Theory can help

generate a more fine-grained typology of therianthropes, moving beyond a single, monolithic "shamanism" explanation to differentiate between representations of stable supernatural entities and representations of dynamic ritual processes.

Table 2: Summary of Interpretations and Cognitive Analyses of Key Therianthropic Figures

Figure	Archaeological Context	Traditional Interpretations (Content-focused)	Cognitive Linguistic Analysis (Process-focused)
Löwenmensch	~40,000 ya; mammoth ivory; high-effort creation; found in an inner chamber.	A deity, an ancestor, a totem, a shaman in lion form.	A stable, double-scope blend of and frames, creating a new ontological category with emergent supernatural properties. A cognitive artifact anchoring a shared concept.
"The Sorcerer"	~13,000 ya; deep, inaccessible cave sanctuary; composite of multiple animals.	"Master of Animals," a costumed shaman performing a ritual, a hunting god.	A dynamic, multi-input blend representing a transient state of transformation, metaphorically accumulating the powers of various animal inputs into a single ritual agent.

IV. A Critical Lens: Epistemological Challenges in Cognitive Archaeology

While CMT and Blending Theory provide a powerful analytical vocabulary, their application to the deep past is not without significant epistemological challenges. Moving from identifying the *structure* of a cognitive operation to inferring its *content* requires a leap of faith across the cognitive chasm. This section critically examines the two most significant dangers: anachronism and the assumption of universality.

4.1. The Peril of Anachronism: Projecting Modern Minds onto the Paleolithic Past

Anachronism is a chronological inconsistency, and in the context of cognitive archaeology, its most insidious form is the unintentional projection of modern concepts, values, and modes of thought onto past peoples.[64] The discipline must constantly guard against the implicit assumption that Paleolithic minds were simply "us in animal skins," possessing the same cognitive and conceptual furniture but in a different setting.[13]

This problem is particularly acute for theories grounded in embodiment. CMT's central claim is that abstract thought is structured by universal bodily experiences.[23] However, while the basic hardware of the human body has remained stable, the lived, phenomenological reality of that body—the *salience* of its experiences—has changed dramatically. The daily embodied reality of an Ice Age hunter-gatherer was one of constant exposure to cold, a different kinesthetic awareness derived from crafting flint tools and tracking animals over vast landscapes, and a sensory world dominated by natural sounds and smells. This profound difference in lived experience could have led to a different set of primary metaphors or, more subtly, a different weighting and hierarchy among them. For instance, while the schema is basic, the specific containers that a Paleolithic person experienced most frequently—animal bladders, woven baskets, rock shelters, the cave itself—are vastly different from the plastic tubs, glass jars, and rectilinear rooms that dominate modern experience. These experiential differences could alter the entailments and metaphorical extensions of the schema in ways we cannot easily reconstruct.

4.2. Beyond Universals: Cultural Variation and the Limits of Embodied Experience

The most potent challenge to a naive universalism in CMT comes from cross-cultural

linguistics. The Aymara people of the Andean highlands provide a striking counterexample to what was once thought to be a universal conceptual metaphor for time.[67] In most of the world's languages, the future is metaphorically located *in front* of the speaker, and the past is *behind*, grounded in the embodied experience of forward motion ().[69] The Aymara, however, systematically reverse this. Their language and gestures place the past (*nayra pacha*, literally "front time") in front of them, and the future (*qhipa pacha*, "back time") behind them.[71] The underlying logic is not based on motion, but on sight: the past is known and has been "seen," so it lies in the field of vision, in front. The future is unknown and cannot be seen, so it lies behind one's back.[68]

The Aymara case is a methodological bombshell for cognitive archaeology. It demonstrates that culture acts as a powerful filter, selecting and shaping which aspects of embodied experience become the basis for metaphorical mapping. The link is not deterministic; it is mediated by cultural values—in this case, the epistemological value placed on 'seen' versus 'unseen' knowledge. This forces a crucial refinement of the theory, from a "strong embodied universalism" (the body dictates the metaphor) to a "weak embodied universalism" (the body provides a *range* of possible metaphorical grounds, from which culture selects, elaborates, and builds its conceptual systems).[73]

This has profound implications for interpreting Paleolithic art. We cannot assume that the blend of a human and a lion in the *Löwenmensch* would automatically carry the same metaphorical weight for an Aurignacian hunter as it does for us. For a modern observer steeped in Western culture, a lion might primarily metaphorize 'courage,' 'nobility,' or 'royalty'—concepts derived from millennia of specific cultural history, from heraldry to children's fables. For an Ice Age hunter, the cave lion was a real and terrifying competitor and predator. The salient source domain concepts might have been about raw power, danger, the spirit of the hunt, or a specific mythological narrative to which we have absolutely no access.[75] The source domains are always culturally loaded.

This critique, however, does not invalidate the use of these theories. Instead, it leads to a more refined and defensible application. A distinction must be drawn between the *cognitive operation* (the formal mechanics of thought) and the *semantic content* (the specific cultural meaning). We can be far more confident in identifying the cognitive operation—e.g., the use of double-scope blending to create an impossible entity—than in specifying the particular meaning of that entity. The ability to merge the conflicting frames of 'human' and 'lion' is a powerful statement about cognitive capacity. The claim that "The creators of the *Löwenmensch* possessed the capacity for double-scope conceptual integration" is a strong, archaeologically-grounded hypothesis. The claim that "The *Löwenmensch* meant X" is, by contrast, a weak and largely unfalsifiable speculation. This crucial distinction saves the theories from being mere interpretive "just-so stories" and preserves their scientific utility.

V. The Embodied Canvas: Space, Sensation, and Meaning-Making in the Cave

A full cognitive archaeological analysis must re-integrate the mental act of creation with its physical, sensory environment. Drawing on the principles of embodied cognition, which hold that cognition is shaped by the body's interactions with its environment [55], it becomes clear that the cave was not a neutral backdrop but an active participant in the meaning-making process. The meaning of the art was inseparable from the experience of the place.

The Journey as Ritual

Many of the most significant and complex artworks, such as "The Sorcerer" at Les Trois Frères, are not found near cave entrances but are located deep within the subterranean world, in chambers that are difficult, confusing, and sometimes dangerous to access.[58] The physical journey to reach these images—crawling through narrow passages, climbing rock faces, navigating in absolute darkness with flickering torches—would have been a profoundly transformative process. This physical ordeal, involving sensory deprivation, disorientation, and exertion, could have served as a ritualistic primer, inducing altered states of consciousness that prepared the initiate for a powerful encounter with the sacred imagery.[4] The meaning was not just in the destination, but in the embodied act of getting there.

Archeoacoustics and Multimodal Meaning

Recent research in archeoacoustics has revealed a strong, non-random correlation between the placement of Paleolithic art and the acoustic properties of the cave chambers.[82] Paintings and engravings are often clustered in "acoustic hot spots," locations where sound reverberates strongly and echoes are pronounced.[84] This compelling evidence suggests that the art was part of a multimodal experience, one that integrated the visual with the auditory. The images were likely not viewed in silence but were activated by chanting, storytelling, or music, with the cave's natural resonance amplifying the sound. In this context, the art is better understood not as a static "text" to be read, but as a "script" for a dynamic performance. The images may have served as visual cues or focal points for rituals that were primarily auditory and experiential. This reframes the art as a piece of cognitive technology for structuring and facilitating an embodied event, rather than as a standalone repository of symbolic content.

The cave wall was not merely a canvas; it was a resonant membrane between worlds.[4]

The Cave as a CONTAINER Schema

From an embodied cognitive perspective, the cave itself is a powerful, real-world instantiation of the fundamental image schema. The act of entering the cave is a literal enactment of the schema, a physical transition from the outside world of light and life to an inner, hidden, and profoundly different world. This potent physical experience of moving from an outer, profane space to an inner, sacred space could have provided the primary metaphorical basis for conceptualizing the cave as a gateway to a spirit world, the underworld, or a realm of origins.[4] The art within this container is not just incidentally located *in* the cave; its meaning is fundamentally structured *by* the experience of containment. The walls themselves could be seen as a membrane between worlds, and the act of painting as a way to draw power from that other world or to communicate with its inhabitants.[4]

This embodied perspective offers a way to ground the shamanic interpretation in something more than ethnographic analogy. While we cannot know the specific content of Paleolithic beliefs, the universal human neurological response to sensory deprivation, disorientation, and resonant sound can induce altered states of consciousness.[4] The deep cave environment is a natural "technology" for producing such states. Therefore, interpreting therianthropic images as representations of visions experienced under these conditions is not simply projecting a modern cultural practice onto the past. It is a testable hypothesis grounded in the interaction between a modern human nervous system and the specific, reconstructed physical and sensory environment of the Paleolithic artist and audience.

VI. Conclusion: Synthesizing Insights and Charting Future Trajectories

Conceptual Metaphor Theory and Conceptual Integration Theory provide an invaluable theoretical toolkit for the cognitive archaeology of Paleolithic art. They allow us to move beyond descriptive or purely speculative interpretations to build formal models of the cognitive operations underlying the creation of symbolic imagery. By focusing on the *process* of meaning-making, they reframe the central question from the unanswerable "What did this art mean?" to the more tractable "By what cognitive operations could this meaning have been made?". They provide a rigorous, process-oriented language for modeling the creation of

non-real, symbolic concepts like the therianthropes of the Ice Age.

However, this report has demonstrated that their application is fraught with epistemological peril. The profound risk of anachronism and the powerful evidence for cultural specificity in metaphorical mapping demand extreme caution. We must resist the temptation to use these theories to "read" the art directly, as if it were a universal language. The source domains that structure metaphorical thought are laden with cultural values and experiences that are lost to prehistory. The theories are best used as a grammar, not a dictionary; they can illuminate the structure of thought, but not its specific content.

The true power of these cognitive models, then, lies in their ability to generate specific, falsifiable hypotheses that can be investigated with the emerging methodologies of a more scientific archaeology. The future of the field lies not in further interpretive speculation, but in the rigorous testing of these cognitive models against the material record. Several key research trajectories are paramount:

- **Systematic Analysis and Digital Methods:** The ad-hoc analysis of individual figures must be supplemented by the creation of comprehensive digital archives of composite figures like therianthropes. This would allow for systematic, quantitative analysis of their features, identifying statistical patterns in which animal parts are combined with which human parts, their frequency, and their archaeological contexts. Such a database would move the study from anecdotal observation to data-driven analysis.[85]
- **GIS and Spatial Analysis:** The qualitative observation that art is often in "special" places must be quantified. Geographic Information Systems (GIS) can be used to systematically map the placement of different types of art (e.g., therianthropes vs. naturalistic animals vs. geometric signs) in relation to detailed 3D models of cave topography, passage accessibility, and measured acoustic properties. This would allow for the statistical testing of hypotheses about the relationship between image type, embodied experience, and sensory context.[89]
- **Experimental Archaeology and Neuroarchaeology:** While we cannot scan the brains of Paleolithic artists, we can use modern neuroimaging techniques (such as fMRI) on contemporary subjects engaged in analogous tasks. Experiments could investigate the neural networks involved in creating composite imagery from memory, or making art under conditions of sensory deprivation and auditory stimulation. Such studies can provide empirical data on the neural correlates of the cognitive processes—like conceptual blending—hypothesized to be at work.[54]
- **Immersive Virtual Reality (VR):** The embodied experience of the cave, so crucial to its meaning, can be empirically investigated. Immersive VR can be used to create high-fidelity reconstructions of Paleolithic cave environments. Researchers can then conduct controlled psychological experiments within these virtual caves, testing hypotheses about the phenomenological experience of navigating them and encountering the art. This can provide empirical data on the psychological impact of the "embodied canvas" and its role in shaping perception and interpretation.[97]

Conceptual Blending and Metaphor Theory do not provide a direct window into the minds on the wall. To believe they could would be to fall prey to the very anachronism we seek to avoid. Instead, they provide us with a high-resolution map of the cognitive territory we must explore. By combining their theoretical power with the empirical rigor of these emerging archaeological sciences, we can begin to chart that territory with ever-increasing confidence. This synthesis promises to transform speculation about the past into a genuine science of the ancient mind.

Works cited

1. Paleolithic art, an introduction - Smarthistory, accessed on October 13, 2025, https://smarthistory.org/paleolithic-art-an-introduction/
2. "Paleolithic Art: Cognitive Aspects of Human Evolution" - ORIAS, accessed on October 13, 2025, https://orias.berkeley.edu/sites/default/files/2009-summary-gill.pdf
3. What is Cognitive Archaeology? - Cambridge University Press, accessed on October 13, 2025, https://www.cambridge.org/core/services/aop-cambridge-core/content/view/A655C7A8EE8D8997125011E246A463EB/S095977430000086Xa.pdf/what-is-cognitive-archaeology.pdf
4. Re-Thinking Early Prehistoric Art as a Cognitive Technology: - The Italian Academy, accessed on October 13, 2025, https://italianacademy.columbia.edu/sites/default/files/content/Boric-Italian%20Academy-seminar%20Fall%202017%20(1).pdf
5. Reconstructed glacier area and volume changes in the European Alps since the Little Ice Age - TC, accessed on October 13, 2025, https://tc.copernicus.org/articles/19/753/2025/
6. The variable European Little Ice Age - Universität Bern, accessed on October 13, 2025, https://www.hist.unibe.ch/e11168/e52524/e69145/e186327/e1458009/L-Wanner-Pfister-Neukom-Little-ice-Age-22_komp_ger.pdf
7. Estimating the Evolution of a Post-Little Ice Age Deglaciated Alpine Valley through the DEM of Difference (DoD) - MDPI, accessed on October 13, 2025, https://www.mdpi.com/2072-4292/15/12/3190
8. Human population dynamics in Europe over the Last Glacial Maximum - PNAS, accessed on October 13, 2025, https://www.pnas.org/doi/10.1073/pnas.1503784112
9. Climate Change Probably Drove Ice Age Western Europeans To Extinction | IFLScience, accessed on October 13, 2025, https://www.iflscience.com/climate-change-probably-drove-ice-age-western-europeans-to-extinction-75739
10. Interpreting Prehistoric Art: Theories and Challenges | Art History I – Prehistory to Middle Ages Class Notes | Fiveable, accessed on October 13, 2025, https://fiveable.me/art-prehistoric-to-middle-ages/unit-1/interpreting-prehistoric

-art-theories-challenges/study-guide/qAt82bVcAadwClwF
11. Looking for Biological Meaning in Cave Art | American Scientist, accessed on October 13, 2025, https://www.americanscientist.org/article/looking-for-biological-meaning-in-cave-art
12. Prehistoric Art and Human Cognition - World of Paleoanthropology, accessed on October 13, 2025, https://worldofpaleoanthropology.org/2023/04/15/prehistoric-art-and-human-cognition/
13. Cognitive archaeology - Wikipedia, accessed on October 13, 2025, https://en.wikipedia.org/wiki/Cognitive_archaeology
14. Interpretation of the Visual Order of Animal Figures in Paleolithic Cave Paintings in Europe, accessed on October 13, 2025, https://francis-press.com/papers/13786
15. Why Did Prehistoric People Draw in the Caves? - Google Arts & Culture, accessed on October 13, 2025, https://artsandculture.google.com/story/why-did-prehistoric-people-draw-in-the-caves-grotte-chauvet/TwVhKr1gnrB2LQ?hl=en
16. Art and Shamanism: From Cave Painting to the White Cube - MDPI, accessed on October 13, 2025, https://www.mdpi.com/2077-1444/10/1/54
17. Cognitive Archaeology Degree Plan | Interdisciplinary Studies - UCCS, accessed on October 13, 2025, https://inds.uccs.edu/cca
18. Cognitive archaeology, and the psychological assessment of extinct minds - PubMed, accessed on October 13, 2025, https://pubmed.ncbi.nlm.nih.gov/38289186/
19. Cognitive Archaeology, accessed on October 13, 2025, https://core.tdar.org/collection/29509/cognitive-archaeology
20. From Things to Thinking: Cognitive Archaeology Abstract 1. Introduction - ResearchGate, accessed on October 13, 2025, https://www.researchgate.net/profile/Anton-Killin/publication/330727240_From_Things_to_Thinking_Cognitive_Archaeology/links/5c5a2e0f299bf1d14caf29c1/From-Things-to-Thinking-Cognitive-Archaeology.pdf
21. Conceptual Metaphor Theory | SBL Handbook of Style, accessed on October 13, 2025, https://sblhs2.com/2016/11/24/conceptual-metaphor-theory/
22. Metaphors We Live By: George Lakoff and Mark Johnson - YouTube, accessed on October 13, 2025, https://www.youtube.com/watch?v=lYcQcwUfo8c
23. (PDF) Conceptual metaphor theory - ResearchGate, accessed on October 13, 2025, https://www.researchgate.net/publication/311403391_Conceptual_metaphor_theory
24. Conceptual metaphor - Wikipedia, accessed on October 13, 2025, https://en.wikipedia.org/wiki/Conceptual_metaphor
25. Conceptual Metaphor Theory - UCSB MAT, accessed on October 13, 2025, https://www.mat.ucsb.edu/~g.legrady/academic/courses/09f200a/pdf/theory_metaphor.pdf
26. Image schema - Wikipedia, accessed on October 13, 2025,

https://en.wikipedia.org/wiki/Image_schema
27. 5 Cognitive Linguistics II: Image Schemata and Conceptual Metonymy, accessed on October 13, 2025, https://www.austriaca.at/0xc1aa5572%200x003bae54.pdf
28. www.tandfonline.com, accessed on October 13, 2025, https://www.tandfonline.com/doi/full/10.1080/13875868.2024.2377284#:~:text=In%20a%20wider%20sense%2C%20image,of%20objects%20in%20their%20environment.
29. Full article: The Diagrammatic Image Schema Language (DISL), accessed on October 13, 2025, https://www.tandfonline.com/doi/full/10.1080/13875868.2024.2377284
30. Image Schemas and Embodied Cognition | Language and Cognition Class Notes - Fiveable, accessed on October 13, 2025, https://fiveable.me/language-cognition/unit-12/image-schemas-embodied-cognition/study-guide/WRlsAaoBRZW7xq6Q
31. (PDF) On defining image schemas - ResearchGate, accessed on October 13, 2025, https://www.researchgate.net/publication/269931714_On_defining_image_schemas
32. Primary Metaphor and Subjective Experience, accessed on October 13, 2025, https://www.flf.vu.lt/dokumentai/_dokumentai/Lakoff_Johnson_embodied_mind_metaphor.pdf
33. Primary metaphors as a tool for cross-linguistic research Conceptual metaphor has traditionally been defined within Cognitive Li, accessed on October 13, 2025, http://linguistics.berkeley.edu/fillmorefest/ffr/abs_mMUva3.pdf
34. About primary metaphors - SciELO, accessed on October 13, 2025, https://www.scielo.br/j/delta/a/GjXZ4hzfQLqyW6HTZp6ntxG/?lang=en
35. Foundations of Meaning: Primary Metaphors and Primary Scenes by Joseph Edward Grady - eScholarship.org, accessed on October 13, 2025, https://escholarship.org/content/qt3g9427m2/qt3g9427m2_noSplash_87927e5c56786241d4df549fe0a0b4b4.pdf
36. Cognitive linguistics and metaphor research: past successes, skeptical questions, future challenges - SciELO, accessed on October 13, 2025, https://www.scielo.br/j/delta/a/hdcJvfsYDxFBpkFZQVfK4nB/?lang=en
37. Primary metaphor - Wikipedia, accessed on October 13, 2025, https://en.wikipedia.org/wiki/Primary_metaphor
38. Conceptual blending - Wikipedia, accessed on October 13, 2025, https://en.wikipedia.org/wiki/Conceptual_blending
39. Cristóbal Pagán Cánovas - Blending or conceptual integration - Google Sites, accessed on October 13, 2025, https://sites.google.com/site/cristobalpagancanovas/research/blending
40. Conceptual Integration - Mark Turner, accessed on October 13, 2025, https://markturner.org/cinLEA.pdf
41. THE NOTION OF CONCEPTUAL INTEGRATION THEORY - Zien Journals Publishing, accessed on October 13, 2025, https://zienjournals.com/index.php/tjm/article/download/3079/2570/3058

42. Conceptual Integration and Fictive Interaction | Literary Universals Project, accessed on October 13, 2025, https://literary-universals.uconn.edu/2016/09/15/conceptual-integration/
43. Double-scope blending of associative and causal event structures in narratives, accessed on October 13, 2025, https://www.press.uni.lodz.pl/index.php/wul/catalog/download/351/1586/746?inline=1
44. Art for art's sake: Ahuja on Mithen - CogWeb, accessed on October 13, 2025, http://cogweb.ucla.edu/Abstracts/Ahuja_on_Mithen_98.html
45. the prehistory of the mind: stephen mithen, accessed on October 13, 2025, https://glotta.ntua.gr/StateoftheArt/Motor-Action/3martin.htm
46. The Early Prehistory of Human Social Behaviour: Issues of Archaeological Inference and Cognitive Evolution - The British Academy, accessed on October 13, 2025, https://www.thebritishacademy.ac.uk/documents/3952/88p145.pdf
47. Lion Man of Hohlenstein-Stadel - (Intro to Humanities) - Vocab, Definition, Explanations | Fiveable, accessed on October 13, 2025, https://fiveable.me/key-terms/introduction-humanities/lion-man-of-hohlenstein-stadel
48. The Lion Man: an Ice Age masterpiece | British Museum, accessed on October 13, 2025, https://www.britishmuseum.org/blog/lion-man-ice-age-masterpiece
49. The Lion Man of Hohlenstein | Meer, accessed on October 13, 2025, https://www.meer.com/en/89049-the-lion-man-of-hohlenstein
50. Lion-man - Wikipedia, accessed on October 13, 2025, https://en.wikipedia.org/wiki/Lion-man
51. Lion Man of Hohlenstein Stadel: Carving: ArtsLookUp.com, accessed on October 13, 2025, https://www.artslookup.com/prehistoric/lion-man-of-hohlenstein-stadel.html
52. Conceptual Blending, Metaphors, and the Construction of Meaning in Ice Age Europe: An Inquiry Into the Viability of Applying - eScholarship, accessed on October 13, 2025, https://escholarship.org/content/qt64x8g354/qt64x8g354_noSplash_c13833a331442b2695523c0023bfeea7.pdf
53. Hohlenstein-Stadel and the Evolution of Human Conceptual Thought | Cognitive Archaeology, accessed on October 13, 2025, https://cognitivearchaeologyblog.wordpress.com/wp-content/uploads/2017/04/wynn-t-coolidge-f-l-bright-m-2009-hohlenstein-stadel-and-the-evolution-of-human-conceptual-thought-cambridge-archaeological-journal-19-73-83.pdf
54. Cognitive Models in Palaeolithic Archaeology | Oxford Academic, accessed on October 13, 2025, https://academic.oup.com/book/1902
55. Embodied and Extended Numerical Cognition - CUNY, accessed on October 13, 2025, https://files.commons.gc.cuny.edu/wp-content/blogs.dir/1461/files/2021/06/Johnson_Everett_Explorations_in_Archaeology_and_Philosophy_-_Chapter_8_Embodied_andC2A0Extended_Numerical_Cognition.pdf
56. On Tools Making Minds: an Archaeological Perspective on Human Cognitive

Evolution, accessed on October 13, 2025, https://www.researchgate.net/publication/329443794_On_Tools_Making_Minds_an_Archaeological_Perspective_on_Human_Cognitive_Evolution
57. The Sorcerer (cave art) - Wikipedia, accessed on October 13, 2025, https://en.wikipedia.org/wiki/The_Sorcerer_(cave_art)
58. "The Sorcerer" from Trois-Frères Cave, Montesquieu-Avantès, France : History of Information, accessed on October 13, 2025, https://www.historyofinformation.com/detail.php?id=2802
59. Les Trois-Frères - Official website of Cavernes du Volp, accessed on October 13, 2025, https://cavernesduvolp.com/en/grottedestroisfreres
60. Prehistoric sorcerers and postmodern furries: Anthropological point of view - Academic Journals, accessed on October 13, 2025, https://academicjournals.org/journal/IJSA/article-full-text-pdf/17D60B05901
61. The strange and controversial prehistoric 'sorcerer' of the Cave of the Trois Frères, accessed on October 13, 2025, https://www.labrujulaverde.com/en/2020/06/the-strange-and-controversial-prehistoric-sorcerer-of-the-cave-of-the-trois-freres/
62. The Sorcerer of Les Trois Frere, accessed on October 13, 2025, https://www.faculty.umb.edu/gary_zabel/Courses/Phil%20281/Philosophy%20of%20Magic/My%20Documents/Therianthropes.htm
63. The Sorcerer of Chauvet Cave | The Magician - WordPress.com, accessed on October 13, 2025, https://artofthemagician.wordpress.com/2010/06/18/the-sorcerer-of-chauvet-cave/
64. Archaeology and contemporaneity | Archaeological Dialogues ..., accessed on October 13, 2025, https://www.cambridge.org/core/journals/archaeological-dialogues/article/archaeology-and-contemporaneity/37248D1F4365DF055A16D5E8327F668D
65. Anachronism - Wikipedia, accessed on October 13, 2025, https://en.wikipedia.org/wiki/Anachronism
66. Embodied cognition - Wikipedia, accessed on October 13, 2025, https://en.wikipedia.org/wiki/Embodied_cognition
67. A Unique Perspective on Time | The fascinating Language of Aymara | by Rogus - Medium, accessed on October 13, 2025, https://medium.com/@rogusblogus/a-unique-perspective-on-time-the-fascinating-language-of-aymara-a8b5eb7c074a
68. WITH THE FUTURE BEHIND YOU, AND THE PAST IN FRONT - Austin Kleon, accessed on October 13, 2025, https://austinkleon.com/2006/06/27/with-the-future-behind-you-and-the-past-in-front/
69. The Future Could Actually Be Far Behind You. Here's Why. - Science Alert, accessed on October 13, 2025, https://www.sciencealert.com/the-future-could-actually-be-far-behind-you-heres-why
70. Aymara, time backwards; comorant fishing - Then Do Better, accessed on

October 13, 2025, https://www.thendobetter.com/blog/2019/12/6/aymara-time-backwards-comorant-fishing
71. With the Future Behind Them: Convergent Evidence From Aymara Language and Gesture in the Crosslinguistic Comparison of Spatial Construals of - UC Berkeley Linguistics, accessed on October 13, 2025, https://linguistics.berkeley.edu/~sweetser/NunezSweetser2006.pdf
72. TIL in the Aymara language, the past is considered to be in front while the future is behind. Linguists believe this is a visual conception, as you are able to see what is in front like you are able to know the past, but you are unable see what is behind like you are unable to know the future - Reddit, accessed on October 13, 2025, https://www.reddit.com/r/todayilearned/comments/ge9vra/til_in_the_aymara_language_the_past_is_considered/
73. The Cultural Basis of Conceptual Metaphors: The Case of Emotions ..., accessed on October 13, 2025, https://www.lancaster.ac.uk/fass/events/laelpgconference/papers/v05/Ansah.pdf
74. Applied Linguistics Perspectives on Cross-Cultural Variation in Conceptual Metaphor, accessed on October 13, 2025, https://www.tandfonline.com/doi/pdf/10.1207/S15327868MS1804_1
75. Variation in metaphor 13, accessed on October 13, 2025, https://periodicos.ufsc.br/index.php/desterro/article/download/13482/12365/41528
76. Metaphor in Culture - Cambridge University Press, accessed on October 13, 2025, https://www.cambridge.org/core/books/metaphor-in-culture/818345C1B0F60FC6AD67AE47460095EE
77. (PDF) Introduction: Cultural Variation In Metaphor - ResearchGate, accessed on October 13, 2025, https://www.researchgate.net/publication/233459871_Introduction_Cultural_Variation_In_Metaphor
78. The interplay between cultural models and metaphor understanding: a cross-cultural cognitive perspective - PMC - PubMed Central, accessed on October 13, 2025, https://pmc.ncbi.nlm.nih.gov/articles/PMC12239927/
79. Embodied Cognition and the Archaeology of Mind: A Radical Reassessment | Request PDF, accessed on October 13, 2025, https://www.researchgate.net/publication/343268259_Embodied_cognition_and_the_archaeology_of_mind_a_radical_reassessment
80. Grotte des Trois-Frères - Don's Maps, accessed on October 13, 2025, https://www.donsmaps.com/troisfreres.html
81. Cave of the Trois-Frères - Wikipedia, accessed on October 13, 2025, https://en.wikipedia.org/wiki/Cave_of_the_Trois-Fr%C3%A8res
82. The writing on the wall | MIT News | Massachusetts Institute of Technology, accessed on October 13, 2025, https://news.mit.edu/2018/humans-speak-through-cave-art-0221
83. Cross-Modality Information Transfer: A Hypothesis about the Relationship among

Prehistoric Cave Paintings, Symbolic Thinking, and the Emergence of Language - Frontiers, accessed on October 13, 2025, https://www.frontiersin.org/journals/psychology/articles/10.3389/fpsyg.2018.00115/full
84. Cross-Modality Information Transfer: A Hypothesis about the Relationship among Prehistoric Cave Paintings, Symbolic Thinking, and the Emergence of Language - PMC - PubMed Central, accessed on October 13, 2025, https://pmc.ncbi.nlm.nih.gov/articles/PMC5826056/
85. Stone toolmaking and the evolution of human culture and cognition - PMC - PubMed Central, accessed on October 13, 2025, https://pmc.ncbi.nlm.nih.gov/articles/PMC3049103/
86. Stone Tool Assemblage Richness during the Middle and Early Upper Palaeolithic in France - faculty.washington.edu, accessed on October 13, 2025, http://faculty.washington.edu/grayson/JAS25PALEO.PDF
87. The images of composite beings from Central European Upper Paleolithic.... | Download Scientific Diagram - ResearchGate, accessed on October 13, 2025, https://www.researchgate.net/figure/The-images-of-composite-beings-from-Central-European-Upper-Paleolithic-1-man-with-bird_fig7_364256516
88. (PDF) Therianthropes as human-animal relations: contextualising rock art in anthropological perspective - ResearchGate, accessed on October 13, 2025, https://www.researchgate.net/publication/347048369_Therianthropes_as_human-animal_relations_contextualising_rock_art_in_anthropological_perspective
89. GIS Applications in Rock Art | 16 - Taylor & Francis eBooks, accessed on October 13, 2025, https://www.taylorfrancis.com/chapters/edit/10.4324/9780429321900-16/gis-applications-rock-art-sara-garc%C3%AAs-dionysios-danelatos-rita-ferreira-anast%C3%A1cio-george-nash
90. Spatial Analysis of an Ancient Cave Site - Esri, accessed on October 13, 2025, https://www.esri.com/news/arcuser/1000/cave.html
91. Cave Art in Context: Methods for the Analysis of the Spatial Organization of Cave Sites, accessed on October 13, 2025, https://www.researchgate.net/publication/251355334_Cave_Art_in_Context_Methods_for_the_Analysis_of_the_Spatial_Organization_of_Cave_Sites
92. THE USE OF GIS IN THE SPATIAL ANALYSIS OF AN ARCHAEOLOGICAL CAVE SITE - National Speleological Society, accessed on October 13, 2025, https://legacy.caves.org/pub/journal/PDF/V64/v64n1-Moyes.pdf
93. Giving Context to Cave Art | American Scientist, accessed on October 13, 2025, https://www.americanscientist.org/article/contextualizing-cave-art
94. (PDF) Experimental insights into cognition, motor skills, and artistic expertise in Paleolithic art - ResearchGate, accessed on October 13, 2025, https://www.researchgate.net/publication/382867142_Experimental_insights_into_cognition_motor_skills_and_artistic_expertise_in_Paleolithic_art
95. Cognitive Demands of Lower Paleolithic Toolmaking - PMC - PubMed Central, accessed on October 13, 2025, https://pmc.ncbi.nlm.nih.gov/articles/PMC4398452/

96. iCog - In the mind's eye - University of Sheffield, accessed on October 13, 2025, https://icog.sites.sheffield.ac.uk/blog/in-the-minds-eye
97. Cognitive Archaeology Research Articles - R Discovery, accessed on October 13, 2025, https://discovery.researcher.life/topic/cognitive-archaeology/3812496?page=1&topic_name=Cognitive%20Archaeology
98. Creating art, shaping the mind: a psychological approach to Upper Palaeolithic cave art, accessed on October 13, 2025, https://www.youtube.com/watch?v=9GtUtvcomOI